

Dhananjay Pratap Singh Jamwal

📍 India ✉ djamwal2004@gmail.com ☎ +91-9103347633 in dhananjaypratapsinghjamwal

Education

Thapar Institute of Engineering and Technology, Patiala

2022-2026

B.E. in Computer Science and Engineering

- CGPA: 8.16/10
- **Coursework:** Database Management Systems, Software Engineering, Machine Learning, Data Science, Natural Language Processing, Speech Processing

Experience

Technical Intern

June 2024 - July 2024

Indian Computer Emergency Response Team (CERT-In)

- Completed a technical internship under the supervision of senior scientists at CERT-In.
- Learned key cybersecurity concepts through a curated playlist of instructional videos provided by CERT-In.
- Gained foundational understanding of topics such as machine learning applications in cybersecurity, intrusion detection systems, anomaly detection, and cyber threat classification.

Data Science Intern

June 2023 - July 2023

Nucleon (IIT Alumni Initiative) × Centre for Essential Skills, IIT Jammu

- Successfully completed a structured Data Science Internship Program focused on core concepts and practical applications.
- Learned foundational topics including data preprocessing, exploratory data analysis, model building, and evaluation metrics.
- Gained hands-on exposure to data science workflows using Python-based tools such as Pandas, NumPy, and Scikit-learn.

Projects

Diet Recommendation System



- Built a personalized diet recommendation engine using KNN, Cosine Similarity, and SVD to suggest healthy food options based on user preferences.
Preprocessed recipe datasets and implemented similarity-based filtering for accurate and diverse meal suggestions.
Applied key data science concepts such as feature engineering, dimensionality reduction, and model evaluation using Scikit-learn.
- *Tech Stack: Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook*

Punjabi Sarcasm Detection



- Developed a deep-learning pipeline in Python for binary sarcasm classification on a custom dataset of 8,000 Punjabi sentences (4,000 sarcastic, 4,000 non-sarcastic).
Implemented and compared a TensorFlow/Keras BiLSTM (100-dim embeddings, 128 hidden units) and a Hugging Face PyTorch BERT-base classifier (frozen encoder + classification head), achieving 94.5
- *Tech Stack: Python, TensorFlow/Keras, PyTorch, Hugging Face Transformers, NumPy, Pandas.*

AI Pneumonia Detection Tool



- Developed a deep learning-based pneumonia diagnostic tool using VGG16 CNN, achieving 92 accuracy on X-ray scans. Integrated Grad-CAM heatmaps for model interpretability, TTS alerts, and automated PDF reporting. Trained on Kaggle's Chest X-Ray Pneumonia dataset; key features include AI diagnosis, heatmap visualization, TTS-based results, doctor consultation support, and report generation.
- *Tech Stack: — Python, TensorFlow, Keras, OpenCV, PyQt5, ReportLab*

Technologies

Languages: Python, C, C++, SQL, R, MATLAB, Golang

Tools Jupyter Notebook, MySQL Workbench, Colab

Frameworks and Libraries Numpy, Pandas, Pytorch, Tensorflow, Scikit-Learn